In [1]:
```
# Run this first, before anything else
from google.colab import drive
drive.mount('/content/drive')

import os

BASE = '/content/drive/MyDrive/CLTT_Soccer'
FRAMES_DRIVE = f'{BASE}/frames'
CKPT_DIR     = f'{BASE}/checkpoints'
UCF_DIR      = f'{BASE}/ucf101'

for d in [BASE, FRAMES_DRIVE, CKPT_DIR, UCF_DIR]:
    os.makedirs(d, exist_ok=True)

print("Drive mounted. Directories created.")
```

```
Mounted at /content/drive
Drive mounted. Directories created.
```

In [2]:
```
!pip install -q yt-dlp decord av
```

```
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 182.3/182.3 kB 14.7 MB/s eta 0:00:0
0
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```

In [3]:
```
VIDEO_URL  = "https://www.youtube.com/watch?v=gTmR5PMIIjs"   # ← only thing yo
VIDEO_PATH = f'{BASE}/soccer_asmr.mp4'

if not os.path.exists(VIDEO_PATH):
    !yt-dlp -f "bestvideo[height<=720][ext=mp4]+bestaudio[ext=m4a]/best[height
else:
    print("Video already on Drive, skipping download.")
```

```
Video already on Drive, skipping download.
```

In [4]:
```
LOCAL_VIDEO = '/content/soccer_asmr.mp4'

if not os.path.exists(LOCAL_VIDEO):
    print("Copying video to local SSD...")
    !cp "{VIDEO_PATH}" "{LOCAL_VIDEO}"
    print(f"Done: {os.path.getsize(LOCAL_VIDEO)/1e9:.2f} GB")

print("Video ready.")
```

```
Copying video to local SSD...
Done: 2.15 GB
Video ready.
```

In [5]:
```
import os

os.makedirs('/content/soccer_frames', exist_ok=True)

# Video is already local — ffmpeg reads it sequentially (very fast)
```

```
!ffmpeg -i /content/soccer_asmr.mp4 -vf "fps=5,scale=128:128" -q:v 3 /content/

n = len([f for f in os.listdir('/content/soccer_frames') if f.endswith('.jpg')
print(f"Extracted {n} frames")
```

```
[swscaler @ 0x58ed6fcaaa00] deprecated pixel format used, make sure you did set
range correctly
Extracted 55075 frames
```

In [28]:

```
import torch
import torch.nn as nn
import torch.nn.functional as F
from torch.utils.data import Dataset, DataLoader
import torchvision.transforms as T
import torchvision.models.video as video_models
from PIL import Image
import random, numpy as np
from tqdm import tqdm

# ── Kinetics-400 normalization stats (standard for video models) ──────────
MEAN = [0.43216, 0.39467, 0.37645]
STD  = [0.22803, 0.22145, 0.21699]

# ─────────────────────────────────────────────────────────────────────────────
from PIL import Image
import os, random, torch
from torch.utils.data import Dataset

class CLTTDataset(Dataset):
    def __init__(self, frames_dir, clip_len=8, fps=5,
                 pos_window_sec=2, min_offset_sec=0.8, transform=None):
        self.frames_dir  = frames_dir
        self.clip_len    = clip_len
        self.pos_window  = int(pos_window_sec * fps)   # max offset in frames
        self.min_offset  = max(1, int(min_offset_sec * fps))  # min offset in
        self.transform   = transform
        self.frames      = sorted(f for f in os.listdir(frames_dir)
                                  if f.endswith('.jpg'))
        self.N = len(self.frames)
        print(f"{self.N} frames | {self.N/fps/3600:.2f} hrs | "
              f"pos_window={pos_window_sec}s | min_offset={min_offset_sec}s")

    def _load_clip(self, start):
        imgs = []
        for i in range(self.clip_len):
            idx  = min(start + i, self.N - 1)
            img  = Image.open(f'{self.frames_dir}/{self.frames[idx]}').convert
            imgs.append(self.transform(img) if self.transform else img)
        return torch.stack(imgs, dim=1)   # C, T, H, W

    def __len__(self):
        return max(0, self.N - self.clip_len - self.pos_window - 1)

    def __getitem__(self, idx):
```

```python
        clip_a = self._load_clip(idx)
        offset = random.randint(self.min_offset, self.pos_window)
        clip_b = self._load_clip(min(idx + offset, self.N - self.clip_len - 1))
        return clip_a, clip_b

# ─────────────────────────────────────────────────────────────────────────────
class CLTTModel(nn.Module):
    """R(2+1)D-18 encoder + MLP projection head."""
    def __init__(self, proj_dim=128):
        super().__init__()
        backbone = video_models.r2plus1d_18(weights=None)

        self.stem    = backbone.stem
        self.layer1  = backbone.layer1
        self.layer2  = backbone.layer2
        self.layer3  = backbone.layer3
        self.layer4  = backbone.layer4
        self.avgpool = backbone.avgpool   # → (B, 512, 1, 1, 1)

        self.projector = nn.Sequential(
            nn.Linear(512, 512),
            nn.BatchNorm1d(512),
            nn.ReLU(inplace=True),
            nn.Linear(512, proj_dim),
        )

    def encode(self, x):
        x = self.stem(x)
        x = self.layer1(x)
        x = self.layer2(x)
        x = self.layer3(x)
        x = self.layer4(x)
        x = self.avgpool(x)
        return x.flatten(1)              # (B, 512)

    def forward(self, x):
        z = self.projector(self.encode(x))
        return F.normalize(z, dim=-1)   # (B, proj_dim)

# ─────────────────────────────────────────────────────────────────────────────
class NTXentLoss(nn.Module):
    """NT-Xent (Normalized Temperature-scaled Cross Entropy)."""
    def __init__(self, temperature=0.07):
        super().__init__()
        self.T = temperature

    def forward(self, z_a, z_b):
        B   = z_a.size(0)
        z   = torch.cat([z_a, z_b], dim=0)                      # (2B, D)
        sim = torch.mm(z, z.T) / self.T                         # (2B, 2B)

        mask = torch.eye(2 * B, dtype=torch.bool, device=z.device)
        sim  = sim.masked_fill(mask, float('-inf'))
```

```python
        labels = torch.cat([
            torch.arange(B, 2 * B, device=z.device),
            torch.arange(0, B,     device=z.device),
        ])
        return F.cross_entropy(sim, labels)

print("Classes defined.")
```

```
Classes defined.
```

In [29]:
```python
import torch.optim as optim
from torch.cuda.amp import autocast, GradScaler
import json

def run_pretraining(frames_dir, ckpt_dir, epochs=2, batch_size=32, lr=3e-4):
    device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
    print(f"Device: {device}")

    train_transform = T.Compose([
        T.RandomCrop(112),
        T.RandomHorizontalFlip(),
        T.ColorJitter(brightness=0.4, contrast=0.4, saturation=0.4, hue=0.1),
        T.ToTensor(),
        T.Normalize(MEAN, STD),
    ])

    dataset = CLTTDataset(frames_dir, clip_len=8, fps=5,
                          pos_window_sec=2, min_offset_sec=0.8,
                          transform=train_transform)

    loader = DataLoader(dataset, batch_size=32, shuffle=True,
                        num_workers=4, pin_memory=True, drop_last=True)

    model     = CLTTModel(proj_dim=128).to(device)
    criterion = NTXentLoss(temperature=0.07)
    optimizer = optim.AdamW(model.parameters(), lr=lr, weight_decay=1e-4)
    scheduler = optim.lr_scheduler.CosineAnnealingLR(
        optimizer, T_max=epochs * len(loader))
    scaler = torch.amp.GradScaler('cuda')

    loss_history = {'batch': [], 'epoch': []}

    for epoch in range(epochs):
        model.train()
        running_loss = 0.0
        pbar = tqdm(loader, desc=f'Epoch {epoch+1}/{epochs}')

        for clip_a, clip_b in pbar:
            clip_a, clip_b = clip_a.to(device), clip_b.to(device)

            optimizer.zero_grad()
            with autocast():
                z_a = model(clip_a)
```

```python
                z_b = model(clip_b)
                loss = criterion(z_a, z_b)

            scaler.scale(loss).backward()
            scaler.step(optimizer)
            scaler.update()
            scheduler.step()

            batch_loss = loss.item()
            running_loss += batch_loss
            loss_history['batch'].append(batch_loss)
            pbar.set_postfix(loss=f'{batch_loss:.4f}')

        avg = running_loss / len(loader)
        loss_history['epoch'].append(avg)
        print(f'Epoch {epoch+1} — avg loss: {avg:.4f}')

        ckpt = f'{ckpt_dir}/cltt_epoch{epoch+1}.pt'
        torch.save({'epoch': epoch+1,
                    'model_state_dict': model.state_dict(),
                    'loss': avg}, ckpt)
        print(f'Saved → {ckpt}')

    with open(f'{ckpt_dir}/loss_history.json', 'w') as f:
        json.dump(loss_history, f)
    print("Loss history saved.")

    return model

print("Pretraining function ready.")
```

```
Pretraining function ready.
```

In [30]:

```python
LOCAL_FRAMES = '/content/soccer_frames'

pretrained_model = run_pretraining(
    frames_dir = LOCAL_FRAMES,
    ckpt_dir   = CKPT_DIR,
    epochs     = 2,
    batch_size = 32,
    lr         = 3e-4,
)
```

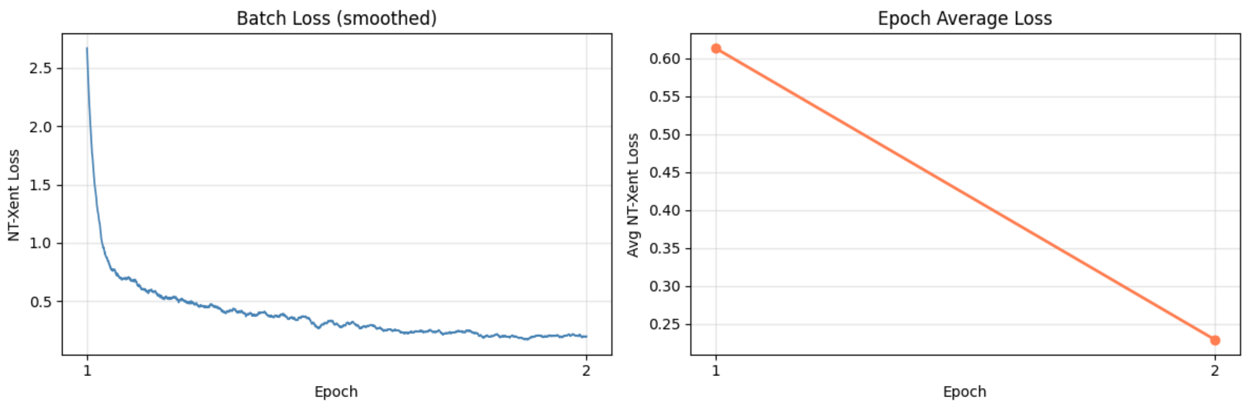
```
Device: cuda
55075 frames | 3.06 hrs | pos_window=2s | min_offset=0.8s
```

```
Epoch 1/2:   0%|          | 0/1720 [00:00<?, ?it/s]/tmp/ipykernel_796/6978338
1.py:42: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated. Pleas
e use `torch.amp.autocast('cuda', args...)` instead.
  with autocast():
/tmp/ipykernel_796/69783381.py:50: UserWarning: Detected call of `lr_schedule
r.step()` before `optimizer.step()`. In PyTorch 1.1.0 and later, you should cal
l them in the opposite order: `optimizer.step()` before `lr_scheduler.step()`.
Failure to do this will result in PyTorch skipping the first value of the learn
ing rate schedule. See more details at https://pytorch.org/docs/stable/optim.ht
ml#how-to-adjust-learning-rate
  scheduler.step()
Epoch 1/2: 100%|██████████| 1720/1720 [09:37<00:00,  2.98it/s, loss=0.4965]
Epoch 1 — avg loss: 0.6132
Saved → /content/drive/MyDrive/CLTT_Soccer/checkpoints/cltt_epoch1.pt
Epoch 2/2: 100%|██████████| 1720/1720 [09:41<00:00,  2.96it/s, loss=0.0591]
Epoch 2 — avg loss: 0.2293
Saved → /content/drive/MyDrive/CLTT_Soccer/checkpoints/cltt_epoch2.pt
Loss history saved.
```

In [31]:

```python
import json, matplotlib.pyplot as plt
import numpy as np

with open(f'{CKPT_DIR}/loss_history.json') as f:
    history = json.load(f)

batch_losses = history['batch']
epoch_losses = history['epoch']
n_epochs     = len(epoch_losses)
epoch_size   = len(batch_losses) // n_epochs

# Smooth batch losses with a rolling average for readability
window  = max(1, epoch_size // 20)
smooth  = np.convolve(batch_losses, np.ones(window) / window, mode='valid')
x_smooth = np.linspace(1, n_epochs, len(smooth))

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

# Left: smoothed batch loss
axes[0].plot(x_smooth, smooth, linewidth=1.2, color='steelblue')
axes[0].set_xlabel('Epoch')
axes[0].set_ylabel('NT-Xent Loss')
axes[0].set_title('Batch Loss (smoothed)')
axes[0].set_xticks(range(1, n_epochs + 1))
axes[0].grid(alpha=0.3)

# Right: per-epoch average
axes[1].plot(range(1, n_epochs + 1), epoch_losses,
             marker='o', linewidth=2, color='coral')
axes[1].set_xlabel('Epoch')
axes[1].set_ylabel('Avg NT-Xent Loss')
axes[1].set_title('Epoch Average Loss')
axes[1].set_xticks(range(1, n_epochs + 1))
axes[1].grid(alpha=0.3)
```

```
plt.tight_layout()
plt.savefig(f'{CKPT_DIR}/loss_curve.png', dpi=150)
plt.show()
print("Plot saved to Drive.")
```



Plot saved to Drive.

In [32]:
```
import shutil

ucf101_dir = f'{UCF_DIR}/UCF-101'

if os.path.exists(ucf101_dir):
    n = sum(len(os.listdir(f'{ucf101_dir}/{c}')) for c in os.listdir(ucf101_di
    print(f"UCF-101 ready — {n} videos across {len(os.listdir(ucf101_dir))} cl
else:
    split_dirs_exist = any(os.path.exists(f'{UCF_DIR}/{s}') for s in ['train',
    if not split_dirs_exist:
        zip_path = f'{UCF_DIR}/archive.zip'
        if not os.path.exists(zip_path):
            raise FileNotFoundError(f"archive.zip not found at {zip_path} — up
        print("Extracting UCF101 from Drive archive.zip...")
        !unzip -oq "{zip_path}" -d "{UCF_DIR}/"

    print("Merging train/test/val into UCF-101/ (one-time, may take a few minu
    os.makedirs(ucf101_dir, exist_ok=True)
    for split in ['train', 'test', 'val']:
        split_dir = f'{UCF_DIR}/{split}'
        if not os.path.exists(split_dir):
            continue
        for cls in os.listdir(split_dir):
            cls_dst = f'{ucf101_dir}/{cls}'
            os.makedirs(cls_dst, exist_ok=True)
            for video in os.listdir(f'{split_dir}/{cls}'):
                src = f'{split_dir}/{cls}/{video}'
                dst = f'{cls_dst}/{video}'
                if not os.path.exists(dst):
                    shutil.move(src, dst)

    n = sum(len(os.listdir(f'{ucf101_dir}/{c}')) for c in os.listdir(ucf101_di
    print(f"Done — {n} videos across {len(os.listdir(ucf101_dir))} classes.")
```

```
UCF-101 ready — 13451 videos across 101 classes.
```

In [33]:
```
splits_dir  = f'{UCF_DIR}/ucfTrainTestlist'
n_splits    = len(os.listdir(splits_dir)) if os.path.exists(splits_dir) else 0

if n_splits < 6:
    drive_zip = f'{UCF_DIR}/splits.zip'
    if os.path.exists(drive_zip):
        print("Found splits.zip on Drive — extracting...")
        !unzip -o "{drive_zip}" -d /tmp/ucf_splits_dir/
    else:
        print("Downloading UCF101 train/test splits from UCF website...")
        !wget -c "https://www.crcv.ucf.edu/data/UCF101/UCF101TrainTestSplits-R
            -O /tmp/ucf_splits.zip
        !unzip -o /tmp/ucf_splits.zip -d /tmp/ucf_splits_dir/
    !cp -rf /tmp/ucf_splits_dir/ucfTrainTestlist "{UCF_DIR}/"
    print(f"Splits ready ({len(os.listdir(splits_dir))} files).")
else:
    print(f"Splits already present ({n_splits} files).")
```

```
Splits already present (7 files).
```

In [34]:
```
import glob
from torchvision.datasets import UCF101

device = torch.device('cuda')

# Load pretrained model from checkpoint if not already in memory
if 'pretrained_model' not in vars() or pretrained_model is None:
    ckpts = sorted(glob.glob(f'{CKPT_DIR}/cltt_epoch*.pt'))
    if not ckpts:
        raise FileNotFoundError(f"No checkpoints in {CKPT_DIR} — run pretraini
    latest = ckpts[-1]
    print(f"Loading checkpoint: {latest}")
    pretrained_model = CLTTModel(proj_dim=128).to(device)
    state = torch.load(latest, map_location=device)
    pretrained_model.load_state_dict(state['model_state_dict'])
    pretrained_model.eval()
    print(f"Loaded epoch {state['epoch']}, loss {state['loss']:.4f}")

def make_ucf_transform():
    return T.Compose([
        T.Resize(128),
        T.CenterCrop(112),
        T.ConvertImageDtype(torch.float32),
        T.Normalize(MEAN, STD),
    ])

def ucf_collate(batch):
    videos = torch.stack([item[0] for item in batch])
    labels = torch.tensor([item[2] for item in batch])
    return videos, None, labels

def load_ucf101(ucf_dir, split='train', fold=1):
```

```python
    return UCF101(
        root              = f'{ucf_dir}/UCF-101',
        annotation_path   = f'{ucf_dir}/ucfTrainTestlist',
        frames_per_clip   = 16,
        step_between_clips = 8,
        fold              = fold,
        train             = (split == 'train'),
        transform         = make_ucf_transform(),
        output_format     = 'TCHW',
    )

@torch.no_grad()
def extract_features(model, loader, device):
    model.eval()
    features, labels = [], []
    for videos, _, lbls in tqdm(loader, desc='Extracting'):
        x = videos.permute(0, 2, 1, 3, 4).to(device)
        h = model.encode(x).cpu()
        features.append(h)
        labels.append(lbls)
    return torch.cat(features), torch.cat(labels)

train_ds = load_ucf101(UCF_DIR, 'train')
test_ds  = load_ucf101(UCF_DIR, 'test')

train_loader = DataLoader(train_ds, batch_size=16, num_workers=4, collate_fn=u
test_loader  = DataLoader(test_ds,  batch_size=16, num_workers=4, collate_fn=u

print("Extracting train features...")
train_feats, train_labels = extract_features(pretrained_model, train_loader, c
print("Extracting test features...")
test_feats, test_labels   = extract_features(pretrained_model, test_loader,  c

print(f"Train: {train_feats.shape}, Test: {test_feats.shape}")
torch.save({'feats': train_feats, 'labels': train_labels}, f'{CKPT_DIR}/train_
torch.save({'feats': test_feats,  'labels': test_labels},  f'{CKPT_DIR}/test_f
print("Features saved.")
```

```
  0%|          | 0/841 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/
torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding
and encoding capabilities of torchvision are deprecated from version 0.22 and w
ill be removed in version 0.24. We recommend that you migrate to TorchCodec, wh
ere we'll consolidate the future decoding/encoding capabilities of PyTorch: htt
ps://github.com/pytorch/torchcodec
  warnings.warn(
100%|██████████| 841/841 [06:22<00:00,  2.20it/s]
  0%|          | 0/841 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/
torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding
and encoding capabilities of torchvision are deprecated from version 0.22 and w
ill be removed in version 0.24. We recommend that you migrate to TorchCodec, wh
ere we'll consolidate the future decoding/encoding capabilities of PyTorch: htt
ps://github.com/pytorch/torchcodec
  warnings.warn(
100%|██████████| 841/841 [06:19<00:00,  2.21it/s]
```

```
Extracting train features...
```

```
Extracting:   0%|          | 0/13099 [00:00<?, ?it/s]/usr/local/lib/python3.12/
dist-packages/torchvision/io/_video_deprecation_warning.py:9: UserWarning: The
video decoding and encoding capabilities of torchvision are deprecated from ver
sion 0.22 and will be removed in version 0.24. We recommend that you migrate to
TorchCodec, where we'll consolidate the future decoding/encoding capabilities o
f PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warni
ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
on are deprecated from version 0.22 and will be removed in version 0.24. We rec
ommend that you migrate to TorchCodec, where we'll consolidate the future decod
ing/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warni
ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
on are deprecated from version 0.22 and will be removed in version 0.24. We rec
ommend that you migrate to TorchCodec, where we'll consolidate the future decod
ing/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warni
ng.py:9: UserWarning: The video decoding and encoding capabilities of torchvisi
on are deprecated from version 0.22 and will be removed in version 0.24. We rec
ommend that you migrate to TorchCodec, where we'll consolidate the future decod
ing/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
/usr/local/lib/python3.12/dist-packages/torchvision/io/video.py:199: UserWarnin
g: The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.
  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 's
ec'.")
Extracting: 100%|██████████| 13099/13099 [1:39:31<00:00,  2.19it/s]
```

```
Extracting test features...
```

```
Extracting:   0%|          | 0/5109 [00:00<?, ?it/s]/usr/local/lib/python3.12/dist-packages/torchvision/io/_video_deprecation_warning.py:9: UserWarning: The video decoding and encoding capabilities of torchvision are deprecated from version 0.22 and will be removed in version 0.24. We recommend that you migrate to TorchCodec, where we'll consolidate the future decoding/encoding capabilities of PyTorch: https://github.com/pytorch/torchcodec
  warnings.warn(
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  warnings.warn("The pts_unit 'pts' gives wrong results. Please use pts_unit 'sec'.")
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Extracting: 100%|██████████| 5109/5109 [39:41<00:00,  2.14it/s]
Train: torch.Size([209582, 512]), Test: torch.Size([81743, 512])
Features saved.
```

In [35]:

```python
from torch.utils.data import TensorDataset

# Load from Drive if needed
train_data = torch.load(f'{CKPT_DIR}/train_feats.pt')
test_data  = torch.load(f'{CKPT_DIR}/test_feats.pt')
train_feats, train_labels = train_data['feats'], train_data['labels']
test_feats,  test_labels  = test_data['feats'],  test_data['labels']
```

```python
NUM_CLASSES = 101

# ── Linear probe: frozen features → trainable linear head ─────────────────
probe = nn.Linear(512, NUM_CLASSES).to(device)
opt   = optim.AdamW(probe.parameters(), lr=1e-3, weight_decay=1e-4)

probe_train = DataLoader(TensorDataset(train_feats, train_labels),
                          batch_size=256, shuffle=True)
probe_test  = DataLoader(TensorDataset(test_feats, test_labels),
                          batch_size=256)

for epoch in range(20):
    probe.train()
    for x, y in probe_train:
        x, y = x.to(device), y.to(device)
        loss  = F.cross_entropy(probe(x), y)
        opt.zero_grad()
        loss.backward()
        opt.step()

    # Eval every 5 epochs
    if (epoch + 1) % 5 == 0:
        probe.eval()
        correct = total = 0
        with torch.no_grad():
            for x, y in probe_test:
                x, y = x.to(device), y.to(device)
                preds   = probe(x).argmax(dim=1)
                correct += (preds == y).sum().item()
                total   += y.size(0)
        print(f"Epoch {epoch+1:2d} — Test Acc: {100*correct/total:.2f}%")
```

```
Epoch  5 — Test Acc: 22.17%
Epoch 10 — Test Acc: 22.09%
Epoch 15 — Test Acc: 22.39%
Epoch 20 — Test Acc: 23.04%
```